

JEE MAIN REVISION MASTER: INVERSE TRIGONOMETRIC FUNCTIONS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 36 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Inverse Trigonometric Functions - Core Application	36 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
-----------------	--

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Inverse Trigonometric Functions - Core Application (Very High Yield — 36 PYQs)

Core Principle: Key problem pattern in Inverse Trigonometric Functions.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let $\alpha = \sin^{-1}(\frac{6}{11})$ and $\beta = \cos^{-1}(\frac{4}{9})$, where inverse trigonometric functions take only the principal value..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "If the domain of the function $f(x) = \sin^{-1}(\frac{1}{x^2-2x-2})$, is $(-\infty, \alpha] \cup [\beta, \gamma] \cup [\delta, \infty)$, then $\alpha + \beta + \gamma + \delta = \dots$ " [Ans: N/A]
- [2026 Evening Shift] Q#2 (MCQ): "If $\sin(\tan^{-1}(x\sqrt{2})) = \cot(\sin^{-1}\sqrt{1-x^2})$, $x \in (0, 1)$, then the value of x is:..." [Ans: A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

Let $\alpha = \sin^{-1}\left(\frac{6}{11}\right)$ and $\beta = \cos^{-1}\left(\frac{4}{9}\right)$, where inverse trigonometric functions take only the principal values. Given below are two statements : Statement I : $\cos(\alpha + \beta) > 0$. Statement II : $\cos(\alpha) < 0$. In the light of the above statements, choose the correct answer from the options given below :

- (A) Both Statement I and Statement II are true
- (B) Both Statement I and Statement II are false
- (C) Statement I is true but Statement II is false
- (D) Statement I is false but Statement II is true

Correct Answer: None

Solution Analysis: $\left(\frac{1}{2}\right) < \left(\frac{6}{11}\right) < \frac{1}{\sqrt{2}} \Rightarrow \sin^{-1}\left(\frac{1}{2}\right) < \sin^{-1}\left(\frac{6}{11}\right) < \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) \dots (i) \Rightarrow 3\sin^{-1}\left(\frac{1}{2}\right) < 3\sin^{-1}\left(\frac{6}{11}\right) < 3\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) \Rightarrow \left(\frac{\pi}{2}\right) < \alpha < \left(\frac{3\pi}{4}\right) \Rightarrow \cos\alpha < 0 \dots (ii)$
 $\left(\frac{4}{9}\right) < \left(\frac{1}{2}\right) < \frac{1}{\sqrt{2}} \Rightarrow \cos^{-1}\left(\frac{4}{9}\right) > \left(\frac{\pi}{3}\right) > \left(\frac{\pi}{4}\right) \Rightarrow 3\cos^{-1}\left(\frac{4}{9}\right) > \pi > \left(\frac{3\pi}{4}\right) \Rightarrow \beta > \pi \dots (iii)$
 Inequalities (i) and (ii) shows the mathematical logic for $\alpha + \beta > \frac{\pi}{2}$